

## ECN 594: Practice Midterm Exam

### Instructions:

- You have **80 minutes**
- You may bring a calculator and notes on a two-sided cheat-sheet (letter-size paper)
- Please be neat. If your work is too messy it will not be graded.
- Be sure to show your working.
- This is a long exam, so there are lots of ways to get points. If you get stuck, move on!
- Good luck!

Name: \_\_\_\_\_

|           |    |    |    |    |       |
|-----------|----|----|----|----|-------|
| Question: | 1  | 2  | 3  | 4  | Total |
| Points:   | 30 | 30 | 20 | 20 | 100   |
| Score:    |    |    |    |    |       |

## 1. Short Answer Questions (30 points)

1. For each question, give a number, True/False/NEI, or a 1-2 sentence answer.
  - (a) (4 points) A monopolist faces demand with elasticity  $\varepsilon = -4$  and has  $MC = \$15$ . What is the profit-maximizing price?
  - (b) (4 points) Consider the logit utility:  $u_{ijt} = \beta \cdot \text{calories}_j + \alpha \cdot p_{jt} + \xi_{jt} + \varepsilon_{ijt}$ . What is the economic interpretation of  $\xi_{jt}$ ?
  - (c) (4 points) True, False, or NEI: In the logit model, if we remove one product from the market, the remaining products' market shares all increase proportionally.
  - (d) (4 points) Why are cost shifters (e.g., input prices) valid instruments for price? Explain both relevance and exogeneity. (1-2 sentences)
  - (e) (4 points) True, False, or NEI: In the logit model, cross-price elasticity between products  $j$  and  $k$  depends on how similar the products are.
  - (f) (3 points) True, False, or NEI: Ramsey pricing sets price equal to marginal cost for all goods.
  - (g) (4 points) A researcher claims that consumer surplus in logit is  $CS = \delta_1 + \delta_2$ . Consider a market with two goods and an outside option. Is this formula correct? Explain.

- (h) (3 points) True, False, or NEI: In a self-selection problem with two types, the high-type's incentive compatibility constraint always binds at the optimum.

## 2. Demand Estimation (30 points)

2. A researcher wants to estimate demand for smartphones using a logit model:

$$u_{ijt} = \beta \cdot \text{screen}_j + \alpha \cdot p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

where  $\text{screen}_j$  is screen size (inches),  $p_{jt}$  is price,  $\xi_{jt}$  is unobserved product quality, and  $\varepsilon_{ijt}$  is i.i.d. Type 1 Extreme Value.

- (a) (5 points) Define the mean utility  $\delta_{jt}$ . Using the Berry inversion, derive an expression for  $\delta_{jt}$  in terms of observed market shares  $s_{jt}$  and the outside option share  $s_{0t}$ .
- (b) (5 points) Write down the linear regression equation the researcher would estimate to recover  $\beta$  and  $\alpha$ . Clearly identify the dependent variable, regressors, and error term.
- (c) (5 points) Why does OLS produce biased estimates of  $\alpha$ ? What is the likely direction of the bias? (2-3 sentences)

- (d) (5 points) A researcher proposes using “cost shifters” (e.g., semiconductor chip prices) as instruments. Give one reason why these instruments might fail the exclusion restriction.
- (e) (5 points) Compared to the basic logit, what is one advantage of adding demographic interactions (e.g., age  $\times$  screen size) to the model?
- (f) (5 points) A phone manufacturer launches two versions of the same phone: “Phone Pro” and “Phone Pro Max,” which are identical except for color. What will the logit model predict about consumer surplus from having both versions available? Is this prediction reasonable? Explain.

### 3. Two-Part Tariffs (20 points)

3. A streaming service charges a monthly subscription fee  $F$  and a per-movie price  $p$ . Marginal cost per movie is  $c = 1$ .

#### Part A: Homogeneous consumers

There are 200 identical consumers, each with demand  $q = 30 - 3p$  for movies per month.

- (a) (5 points) Write down the streaming service's profit as a function of  $F$  and  $p$ . (Hint: What is consumer surplus at price  $p$ ? What is the maximum  $F$  the service can charge?)

- (b) (5 points) Show that the profit-maximizing per-movie price is  $p^* = c = 1$ . (Hint: Take the derivative of profit with respect to  $p$  and set it equal to zero.)

**Part B: Heterogeneous consumers**

Now suppose there are 100 “movie buffs” with demand  $q_H = 20 - 2p$  and 100 “casual viewers” with demand  $q_L = 8 - 2p$ . Marginal cost is  $c = 0$ .

(c) (10 points) The streaming service is considering two pricing strategies:

(a)  $F = 50, p = 0$

(b)  $F = 8, p = 0$

Calculate profit under each strategy. Which is better and why?

#### 4. Self-Selection (20 points)

4. An airline sells two ticket classes: Business (B) and Economy (E). There are two types of travelers: corporate travelers with high willingness to pay and leisure travelers with low willingness to pay.

| Traveler Type            | Willingness to Pay |               |
|--------------------------|--------------------|---------------|
|                          | Business Class     | Economy Class |
| Corporate (50 travelers) | \$800              | \$300         |
| Leisure (100 travelers)  | \$200              | \$150         |

Marginal cost is \$50 for both classes.

- (a) (4 points) If the airline could perfectly identify traveler types, what prices would it charge for each class to each type? What is total profit?
- (b) (4 points) The airline cannot identify types but offers both classes. Write down the incentive compatibility (IC) constraint that ensures corporate travelers buy Business rather than Economy.

(c) (6 points) Suppose the airline sets  $P_B = \$750$  and  $P_E = \$150$ . Explain why these prices are not optimal. Which constraint is violated?

(d) (6 points) Find the profit-maximizing prices for Business and Economy under self-selection. (Hint: Which IC constraint binds? Set it as an equality and solve.)